University of Warsaw
Faculty of Economic Sciences

Machine Learning I

Exam, Date: 2026-05-15 — **Group A**

Timing: 20 minutes Max points: 20
Name and surname: ____________________
ID number: ____________________

Instructions:

- Answer all **20** single-choice questions (**1 pt each**; total **20**).
- Each question has **exactly ONE correct answer**. Mark it by writing the letter (A/B/C/D) in the table below *and* by circling the correct option in the question body.
- To pass you need at least **11 correct answers**. No communication or collaboration.

Answer sheet – write the chosen letter (A/B/C/D) in the cell below each question number.

Q. #	1	2	3	4	5	6	7	8	9	10
Answer										
Q. #	11	12	13	14	15	16	17	18	19	20
Answer										

1. What does a *supervised* learning task require?
   A. Unlabeled inputs only, with no target information
   B. Labeled examples with known input-output pairs
   C. A clustering routine that creates the targets internally
   D. A reinforcement signal from interaction with an environment
2. Which problem is a *regression* task?
   A. Predicting whether an incoming email is spam
   B. Grouping customers into segments without labels
   C. Choosing the next action in a board game
   D. Predicting tomorrow's stock return as a real number
3. Which set is used to *fit* the model parameters?
   A. Training set
   B. Validation set
   C. Test set
   D. Production data
4. The sigmoid function in logistic regression outputs:
   A. The raw linear combination of the input features
   B. A number in $(0, 1)$, interpreted as a probability
   C. The signed distance to the SVM separating margin
   D. An integer class label after applying a threshold
5. Which metric is appropriate for a *regression* problem?
   A. F1-score
   B. ROC AUC
   C. Mean Squared Error
   D. Precision
6. Why can accuracy be misleading on an imbalanced dataset?
   A. A majority-class predictor can still reach high accuracy
   B. The metric is not differentiable on the logit scale
   C. Its value depends heavily on the chosen decision threshold and the calibration of scores
   D. It penalizes false positives much more strongly than false negatives in the score
7. SMOTE balances classes primarily by:
   A. Randomly dropping samples from the majority class
   B. Increasing the L2 regularization strength applied to the classifier during training
   C. Computing class weights for the loss inversely proportional to class frequencies
   D. Synthesizing minority samples between nearest neighbors in feature space
8. As k in K-Nearest Neighbors is increased, the model tends to:
   A. Overfit more and have a higher prediction variance
   B. Produce smoother boundaries with a higher bias
   C. Become more sensitive to outliers in the training data
   D. Ignore the feature scales of the input data
9. In SVM, the regularization parameter C controls:
   A. The width of the Gaussian kernel (RBF spread)
   B. The choice of kernel family (linear, polynomial, RBF, sigmoid) used by the solver
   C. The trade-off between margin width and training errors
   D. The number of support vectors automatically selected during the fitting procedure
10. The main purpose of k-fold cross-validation is to:
    A. Enlarge the effective size of the training dataset used to fit the model parameters
    B. Permanently replace the held-out test set
    C. Make the training procedure deterministic across different runs and hardware setups
    D. Obtain a more reliable estimate of out-of-sample error
11. The *curse of dimensionality* hurts KNN because, in high dimensions:
    A. KNN defaults to Manhattan distance instead of Euclidean
    B. Cosine similarity automatically replaces the Euclidean metric
    C. Nearest-neighbor distances become nearly equal and lose information
    D. Feature scaling matters less than feature centering does
12. How does L1 (Lasso) regularization differ from L2 (Ridge)?
    A. L1 can drive some coefficients exactly to zero, enabling feature selection
    B. L1 shrinks all coefficients proportionally to their magnitude, just as ordinary L2 regression does
    C. L2 yields sparse coefficient vectors by its mathematical construction
    D. L2 typically requires subgradient methods rather than gradients
13. Bagging (Bootstrap Aggregating) primarily aims to:
    A. Increase the bias of each individual base learner via depth restrictions during training
    B. Replace cross-validation with a single bootstrap resampling
    C. Reduce ensemble variance by averaging bootstrap-trained models
    D. Sequentially update training-example residuals via weighted gradient steps after each iteration
14. In an RBF-kernel SVM, the γ parameter controls:
    A. The reach of a single training point's influence on predictions
    B. The same trade-off as captured by the regularization parameter C
    C. The number of support vectors automatically determined by the solver at the fitting step
    D. The loss function family used by the optimizer during the training procedure
15. Stratified k-fold is preferred over plain k-fold when:
    A. The classes are perfectly balanced across both training and validation portions of the data
    B. The task is regression with a continuous numeric target variable on the real line
    C. Classes are imbalanced and proportions should hold per fold
    D. The data has no temporal ordering between observations
16. Nested cross-validation is used primarily to:
    A. Parallelize hyperparameter search and final evaluation across cores
    B. Replace stratification when class proportions vary across folds
    C. Avoid preprocessing the input features inside the inner loop
    D. Give an unbiased generalization estimate after hyperparameter tuning
17. Standard random k-fold CV is inappropriate for time-series forecasting because:
    A. The scikit-learn implementation does not support this case for ordered data series
    B. Random folds mix future data into training, leaking information
    C. Class labels cannot be stratified across consecutive time periods
    D. Mean squared error is undefined on time-ordered observations of equal length samples
18. A classifier whose predicted probability of 0.9 corresponds to only 60% empirical positives is:
    A. Perfectly calibrated and needs no further adjustment
    B. Accurate at all decision thresholds by construction
    C. Affected by class imbalance rather than by miscalibration
    D. Over-confident and can be repaired by Platt scaling or isotonic regression
19. The *kernel trick* in the dual SVM allows:
    A. Computing inner products in a high-dimensional space without an explicit map
    B. Removing the need for support vectors from the dual solution
    C. Turning any non-linear kernel into an equivalent linear one by means of a coordinate change
    D. Replacing the margin formulation with a softmax output layer
20. Concept drift differs from data (covariate) drift because:
    A. The two terms refer to the same underlying phenomenon in production
    B. Data drift changes $P(X)$; concept drift changes the relationship $P(Y \mid X)$
    C. Concept drift can occur only in the training data, never in production
    D. Data drift is not measurable in standard production monitoring setups

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Machine Learning I

Exam, Date: 2026-05-15 — **Group B**

Timing: 20 minutes Max points: 20

Name and surname: ____________________

ID number: ____________________

Instructions:

- Answer all **20** single-choice questions (**1 pt each**; total **20**).
- Each question has **exactly ONE correct answer**. Mark it by writing the letter (A/B/C/D) in the table below *and* by circling the correct option in the question body.
- To pass you need at least **11 correct answers**. No communication or collaboration.

Answer sheet – write the chosen letter (A/B/C/D) in the cell below each question number.

Q. #	1	2	3	4	5	6	7	8	9	10
Answer										
Q. #	11	12	13	14	15	16	17	18	19	20
Answer										

1. A classification task differs from a regression task because:
   A. It is typically trained with a higher learning rate
   B. Its target variable takes categorical values instead of continuous ones
   C. It treats every input feature as a categorical variable that must be one-hot encoded
   D. Its loss surface is convex while regression's is not
2. In ML vocabulary, a *feature* is:
   A. An input variable used to predict the target
   B. A hyperparameter that controls the learning algorithm
   C. A regularization term added to the training loss
   D. A component of the loss function being optimized
3. The *validation* set is primarily used to:
   A. Fit the parameters of the final deployed model
   B. Provide the final unbiased performance estimate at completion
   C. Tune hyperparameters and compare candidate models
   D. Compute the gradient during back-propagation steps
4. The standard training loss of logistic regression is:
   A. Mean squared error
   B. Hinge loss
   C. The 0–1 loss
   D. Binary cross-entropy (log-loss)
5. The F1-score is:
   A. The arithmetic mean of precision and recall
   B. The harmonic mean of precision and recall
   C. Equal to accuracy whenever the classes are balanced
   D. The area under the ROC curve across thresholds
6. Training error low but validation error much higher typically indicates:
   A. Overfitting and high variance
   B. Underfitting and high bias
   C. A perfectly calibrated classifier
   D. A pipeline that is free from leakage
7. One-hot encoding is most appropriate for:
   A. Continuous numeric variables on a comparable scale
   B. Nominal (unordered) categorical variables
   C. Ordinal variables with a clear natural order
   D. Raw time-stamps recorded from a clock
8. Combining oversampling with a cleaning step (e.g. SMOTE followed by Tomek links) is known as:
   A. Random oversampling of the minority class
   B. Class weighting in the training loss
   C. Hybrid resampling (e.g. SMOTE–Tomek / SMOTEENN)
   D. Stacking different resamplers in a pipeline
9. Min–max scaling transforms features to:
   A. Zero mean and unit variance
   B. A bounded range, typically $[0, 1]$
   C. Integer-valued discretization buckets
   D. Rank-based percentile values
10. ROC AUC measures:
    A. The accuracy at the default 0.5 decision threshold across cross-validation folds
    B. The precision when recall is held fixed at 100%
    C. The quality of ranking positives versus negatives across thresholds
    D. The mean squared error of the predicted probabilities relative to true class labels
11. According to the bias–variance trade-off, increasing model flexibility usually:
    A. Raises both bias and variance simultaneously
    B. Lowers both bias and variance simultaneously
    C. Has no relationship with model complexity
    D. Lowers bias while raising variance
12. Which statement about Ridge (L2) regression is correct?
    A. It enforces sparsity in the trained coefficient vector by zeroing small entries
    B. It has no closed-form solution available analytically
    C. It shrinks coefficients toward zero without setting any to zero
    D. It is fully invariant to the scale of the input features used during training
13. In `CalibratedClassifierCV`, setting `method='isotonic'` fits:
    A. A non-parametric piecewise calibration map on classifier scores
    B. A parametric sigmoid map fit on the classifier scores using Newton's method
    C. A linear regression on the predicted class labels, ignoring scores
    D. A Bayesian posterior over the calibration parameters of the fit on training data
14. A soft-margin SVM introduces slack variables in order to:
    A. Eliminate the C hyperparameter from the dual optimization
    B. Simplify the dual problem to a closed-form solution
    C. Replace the kernel function with an explicit feature map
    D. Allow some points inside the margin, traded off against C
15. Grid search vs. random search for hyperparameter tuning:
    A. Grid explores a fixed discrete grid; random samples from a distribution
    B. Random search converges considerably more slowly than grid search in low-dimensional spaces
    C. Grid search bypasses cross-validation when used inside pipelines
    D. Random search produces reproducible results without needing a seed
16. In an imbalanced classification pipeline, resampling (e.g. SMOTE) should be applied:
    A. Before the train/validation split, on the complete combined dataset including held-out points
    B. After the model has already been fit on training
    C. Only when class weighting is unavailable for the chosen estimator implementation
    D. Only within the training fold of each CV split, to avoid leakage
17. Under its standard assumptions, the Gauss–Markov theorem states that OLS is:
    A. The minimum-variance unbiased linear estimator (BLUE)
    B. The maximum-likelihood estimator under Gaussian errors
    C. The optimal estimator even under heteroskedastic errors
    D. A biased but asymptotically efficient linear estimator
18. Recursive Feature Elimination (RFE) selects features by:
    A. Ranking them once by correlation and discarding the lowest
    B. Drawing random subsets of features at each iteration
    C. Iteratively fitting an estimator and removing the least important feature
    D. Adding features one at a time by largest information gain
19. Bayesian hyperparameter search (e.g. `BayesSearchCV`) differs from random search because it:
    A. Performs evaluations completely independently of one another at every iteration step
    B. Skips cross-validation when using a Gaussian-process prior
    C. Has lower per-evaluation cost than random search
    D. Uses a probabilistic surrogate model to propose the next configuration
20. The Brier score for binary classification is:
    A. The arithmetic mean of all predicted probabilities
    B. The mean squared error between predicted probabilities and outcomes
    C. A score numerically identical to the log-loss in every case
    D. A score defined only for perfectly calibrated probabilistic models

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Machine Learning I

Exam, Date: 2026-05-15 — **Group C**

Timing: 20 minutes Max points: 20

Name and surname: ______________________

ID number: ______________________

Instructions:

- Answer all **20** single-choice questions (**1 pt each**; total **20**).
- Each question has **exactly ONE correct answer**. Mark it by writing the letter (A/B/C/D) in the table below *and* by circling the correct option in the question body.
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Answer sheet – write the chosen letter (A/B/C/D) in the cell below each question number.

Q. #	1	2	3	4	5	6	7	8	9	10
Answer										
Q. #	11	12	13	14	15	16	17	18	19	20
Answer										

1. Which of the following is an example of *unsupervised* learning?
   A. Predicting house prices from their input features
   B. Labeling emails as spam versus non-spam using a manually labeled training set
   C. Predicting credit default risk from historical labeled defaults of past customers
   D. Clustering customers by purchase behavior without any labels
2. The *test* set should be used:
   A. During the training loop to update the model weights at every gradient step
   B. To tune hyperparameters during model selection
   C. At the very end, to estimate the generalization error
   D. At every epoch to compute the training gradient
3. Ordinary Least Squares (OLS) minimizes:
   A. The sum of squared residuals
   B. The sum of absolute residuals
   C. The cross-entropy of the predictions
   D. The width of the decision margin
4. Which of the following is a *classification* metric?
   A. R^2
   B. Precision
   C. MAE
   D. MSE
5. An outlier-robust scaler is:
   A. Standard Scaler (z-score normalization)
   B. Min–Max Scaler (linear range mapping)
   C. Robust Scaler (median and IQR scaling)
   D. Log Scaler (logarithmic transformation)
6. Overfitting can be mitigated by:
   A. Increasing the model complexity well beyond the current level on the same training dataset
   B. Removing the validation set during model training
   C. Training for more epochs once the validation loss has plateaued
   D. Adding regularization, using more data, or proper cross-validation
7. *Target encoding* replaces a categorical value with:
   A. A randomly drawn integer sampled from a uniform distribution over the seen categories
   B. A statistic of the target conditional on the category
   C. A one-hot indicator vector across the observed categories
   D. The character length of the category's string label written in the alphabet
8. Which method generates synthetic minority samples by interpolating between minority nearest neighbors?
   A. Random undersampling
   B. Tomek links
   C. SMOTE
   D. Class weighting
9. When using Euclidean distance in KNN, a crucial assumption is that:
   A. The features are ordinal in nature
   B. The target variable takes continuous values
   C. The data is linearly separable in feature space
   D. The features are on comparable numeric scales
10. Precision is defined as:
   A. $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$
   B. $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$
   C. $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$
   D. $(\mathrm{TP}+\mathrm{TN})/\mathrm{total}$
11. The `validation_curve` function in scikit-learn plots:
   A. Training and validation scores as a function of one hyperparameter
   B. Training and validation scores plotted as a function of the training-set size in samples
   C. The cross-validation fold count against the chosen scoring metric
   D. The cumulative score gain from incrementally adding features
12. Elastic Net combines:
   A. Voting and bagging as ensemble methods
   B. Logistic and linear regression in a single model
   C. Ridge and LASSO regression applied in two separate sequential pipeline stages on residuals
   D. L1 and L2 penalties combined in a single regularization objective
13. *Stacking* combines base learners by:
   A. Training a meta-learner on the base models' predictions
   B. Sequentially reweighting examples after each base learner is fit
   C. Choosing the single base estimator with the highest accuracy
   D. Performing a deterministic majority vote with a tie-break rule
14. For a task where false negatives are more costly than false positives, a sensible metric is:
   A. Plain accuracy computed on the held-out test set used for final evaluation
   B. F1, weighting precision and recall equally in the harmonic mean formula
   C. F_β with $\beta > 1$, weighting recall above precision
   D. ROC AUC computed across all decision thresholds
15. In logistic regression, the coefficient β_j is interpreted as:
   A. The direct change in predicted probability per unit of x_j
   B. The change in the log-odds of the positive class per unit of x_j
   C. The slope of the CDF of x_j evaluated at its sample mean
   D. The variance of x_j across the training distribution after preprocessing
16. `TimeSeriesSplit` differs from `KFold` because it:
   A. Shuffles the samples randomly between the folds
   B. Stratifies each fold by the class proportions in the data
   C. Uses bootstrap sampling for each training fold
   D. Uses expanding training windows followed by a future validation block
17. Data leakage occurs when:
   A. Missing values in the training data are imputed during preprocessing before modelling
   B. Information from outside the training set leaks into the model fit
   C. A model is regularized too strongly during training, leading to high training error
   D. Cross-validation is used to estimate model performance
18. Multi-collinearity between features in OLS regression mainly:
   A. Inflates the variance of the estimated coefficients
   B. Reduces the residual sum of squares to zero
   C. Lowers the bias of the estimated coefficients
   D. Forces the design matrix to be exactly rank one in the most extreme correlation case
19. The Wasserstein (earth-mover's) distance is used in ML monitoring to:
   A. Compress and reconstruct production datasets during preprocessing
   B. Perform direct classification of points without a trained model
   C. Quantify how much two distributions differ, useful for drift detection
   D. Replace cross-validation as the main evaluation procedure
20. Isotonic regression, when used as a calibration step:
   A. Fits a monotonic non-parametric mapping from scores to probabilities
   B. Generally outperforms Platt scaling regardless of dataset size
   C. Requires the underlying classifier to be a linear model
   D. Is a parametric sigmoid fit applied to the classifier scores

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Machine Learning I

Exam, Date: 2026-05-15 — **Group D**

Timing: 20 minutes Max points: 20

Name and surname: ______________________

ID number: ______________________

Instructions:

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Answer										
Q. #	11	12	13	14	15	16	17	18	19	20
Answer										

1. The dataset used to fit model parameters is the:
   A. Training set
   B. Validation set
   C. Test set
   D. Production set
2. A *binary* classification problem has:
   A. A continuous-valued target variable
   B. No labeled examples available during training
   C. Exactly two possible class labels
   D. An unbounded continuous output space
3. Mean Absolute Error (MAE) is:
   A. A classification metric for binary outcomes
   B. A quantity numerically equal to MSE in general
   C. A metric undefined when the targets are continuous
   D. The mean of absolute prediction-target differences
4. Feature scaling matters *most* for:
   A. Algorithms that split features at thresholds without computing distances
   B. Models that ignore the input features and learn only the marginal mean of the target variable
   C. Algorithms that compute distances or gradients in input space (KNN, SVM-RBF)
   D. Estimators that simply memorize the training data for inference time
5. Logistic regression predicts:
   A. A separate regression coefficient on each input feature alongside an intercept term
   B. Raw class labels via rule-based logical conditions
   C. Only binary features after applying a threshold
   D. A class probability via sigmoid of a linear combination
6. Recall (sensitivity) is defined as:
   A. $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$
   B. $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$
   C. $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$
   D. $(\mathrm{TP}+\mathrm{FP})/\mathrm{total}$
7. The main reason to use a scikit-learn `Pipeline` is:
   A. To tune hyperparameters automatically across estimators
   B. To speed up inference at deployment time substantially across different request patterns
   C. To chain preprocessing with the estimator and prevent fold leakage
   D. To avoid using cross-validation during model selection
8. Applying z-score normalization to a feature yields values with:
   A. A bounded range like $[0, 1]$ from observed minimum and maximum
   B. An integer percentile rank on a fixed uniform discrete grid
   C. The natural logarithm shifted by one to avoid log of zero
   D. A mean of zero and a variance of one across samples
9. Setting `class_weight='balanced'` in scikit-learn:
   A. Duplicates minority-class samples in the training data
   B. Weights the loss inversely to class frequencies in training
   C. Removes majority-class samples from the training data
   D. Requires SMOTE or another resampler to be applied beforehand inside the pipeline
10. A hard-voting classifier predicts:
    A. The majority class among the base estimators' predictions
    B. The class with the highest average predicted probability across estimators
    C. The label produced by the single most accurate base estimator
    D. The class with weighted votes inversely scaled by class frequencies
11. The dual formulation of SVM is useful mainly when:
    A. The number of features is much smaller than the number of samples
    B. One wants to apply the kernel trick for non-linear boundaries
    C. One wants to skip cross-validation during model selection
    D. Only linear kernels are available in the implementation library
12. Learning curves (training vs validation error against training-set size) can diagnose:
    A. Only the optimal choice of kernel function for the data
    B. Only the effect of the random seed on the training run
    C. Underfitting (errors high and close) versus overfitting (large gap)
    D. Whether stratification is necessary for the cross-validation
13. Variance Threshold feature selection removes features:
    A. Most correlated with the target above a fixed threshold
    B. Whose sample variance falls below a chosen threshold
    C. Selected by a forward stepwise selection procedure applied during preprocessing steps
    D. Containing missing values inside the training data after the imputation step
14. Why is log-loss (binary cross-entropy) a *proper* scoring rule for probabilistic classifiers?
    A. Because it rewards any confident prediction equally
    B. Because its value equals exactly zero for the optimal model evaluated on the training data
    C. Because it ignores the sign of the prediction error
    D. Because it is minimized in expectation by the true conditional probability
15. A fully *calibrated* classifier means that:
    A. Its decision threshold is exactly 0.5 for all classes
    B. Among predictions with probability p, about a fraction p are positive
    C. Its accuracy is maximized over the choice of threshold
    D. Its precision and recall are numerically equal at every decision threshold considered
16. The `cross_val_score` function in scikit-learn returns:
    A. An array of per-fold test scores
    B. The mean test score across the folds as a single float
    C. The estimator fitted on the last training fold of the run
    D. The hyperparameters selected as optimal across the folds
17. Adding a *missing indicator* column next to an imputed feature is useful because:
    A. Imputation introduces a bias that the indicator absorbs
    B. It standardizes the imputed feature column implicitly during scaling
    C. The missingness pattern itself may carry predictive information
    D. It serves as an additional regularization mechanism for the estimator
18. In an RBF-kernel SVM, setting γ very *high* tends to:
    A. Produce smoother decision boundaries and risk underfitting
    B. Limit each training point's influence to a tight neighborhood (overfitting risk)
    C. Become equivalent to a linear kernel in the dual formulation
    D. Increase the margin width proportionally to γ's value
19. Compared with MSE, the Mean Absolute Error (MAE) as a loss function:
    A. Squares each error to emphasize large deviations strongly
    B. Has a smooth, continuous gradient also at zero error
    C. Equals the MSE when residuals are normally distributed
    D. Is more robust to outliers in the target variable
20. In *Platt scaling*, calibration is obtained by:
    A. Fitting a logistic regression on classifier scores against true labels
    B. Replacing the softmax output layer with an argmax operation
    C. Rescaling the input features to zero mean before training the classifier model
    D. Removing the intercept term from the trained classifier